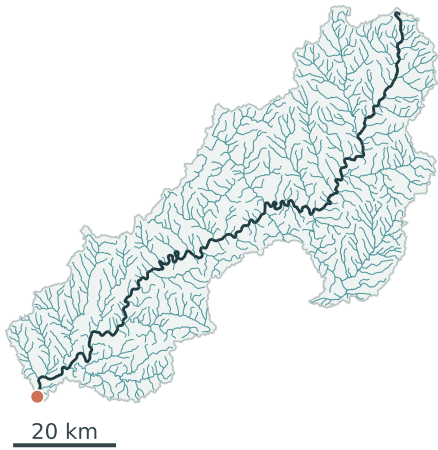


How does river form reorganize the arrival of a DOC signal?

Measured channels and fixed representatives; identical concentration forcing, total flow and normalized velocity.

a Elongated / tributary-rich
03030852



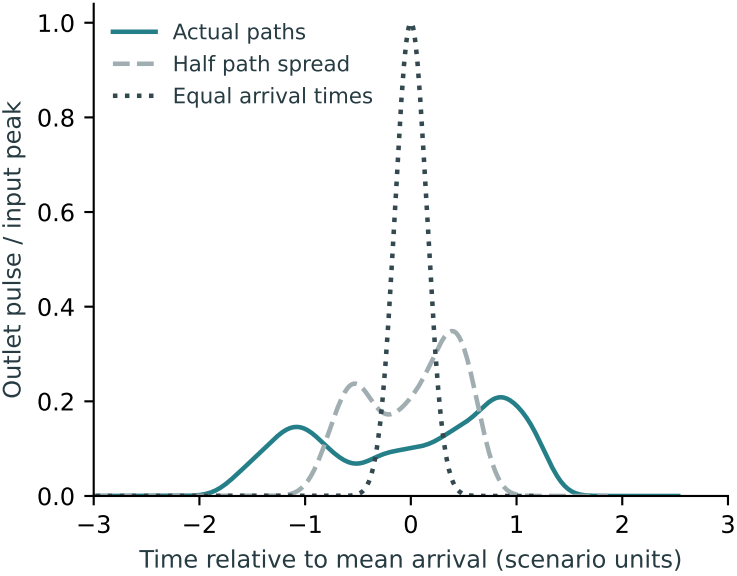
b Mainstem dominated / sparse
06893990



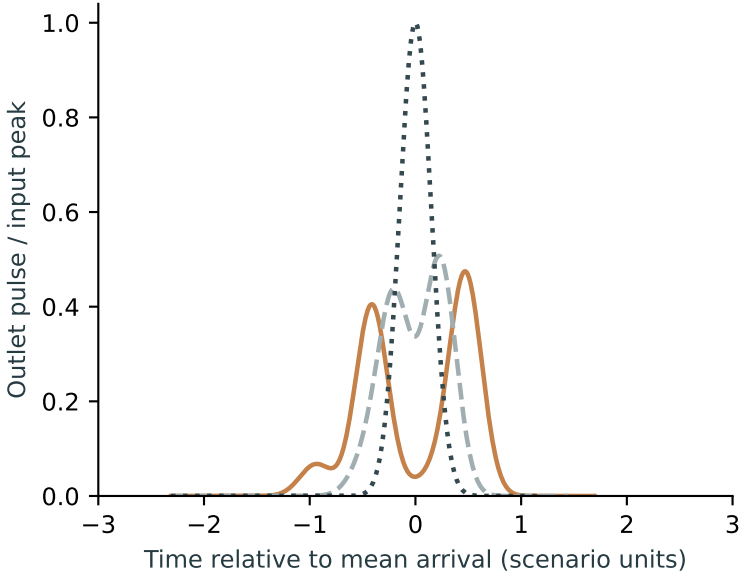
c Broad / tributary-rich
07099200



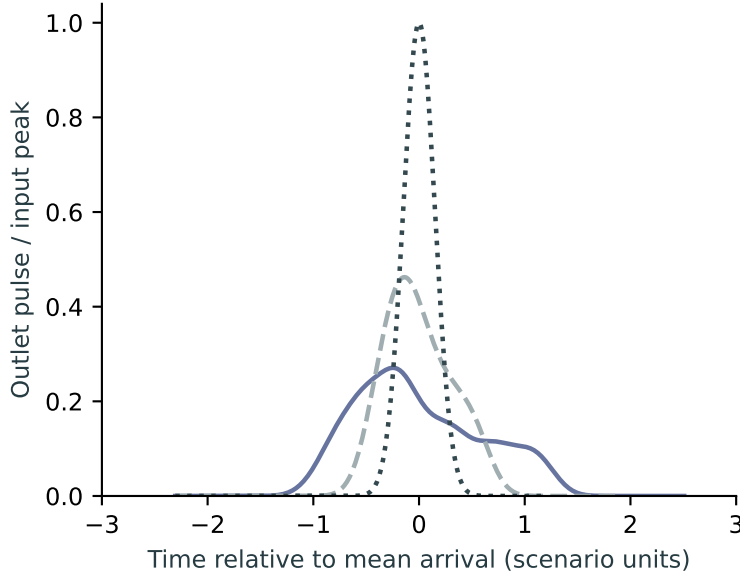
d Within-network path controls



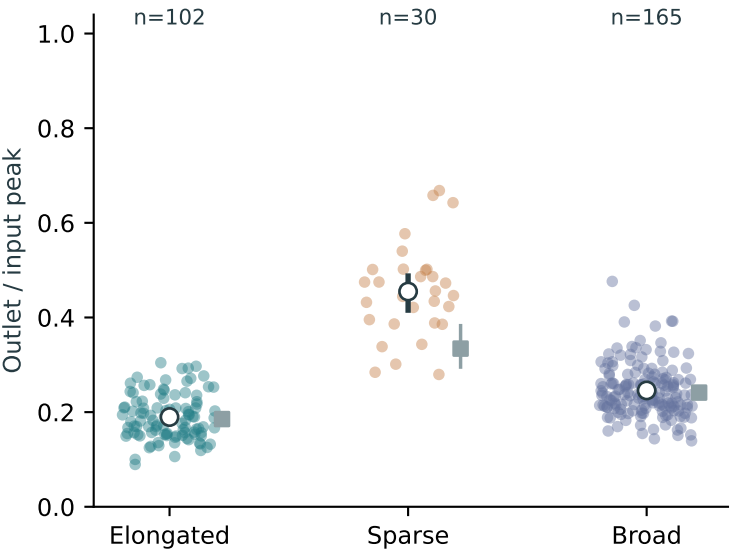
e Within-network path controls



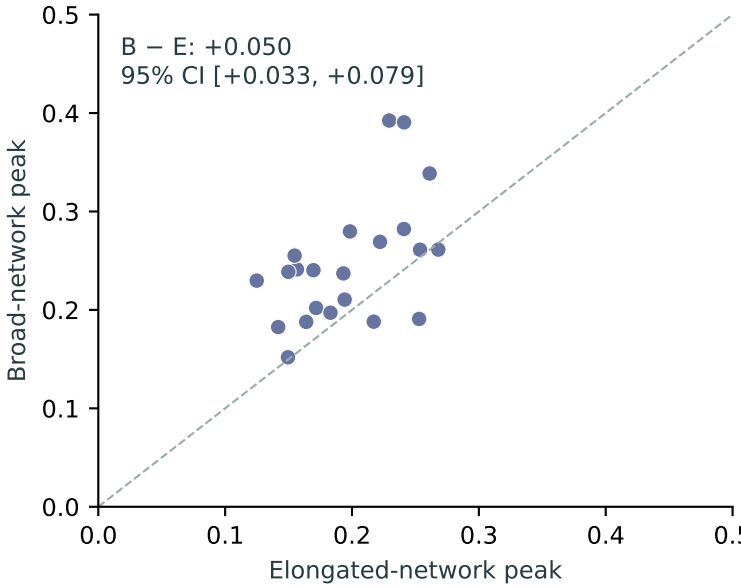
f Within-network path controls



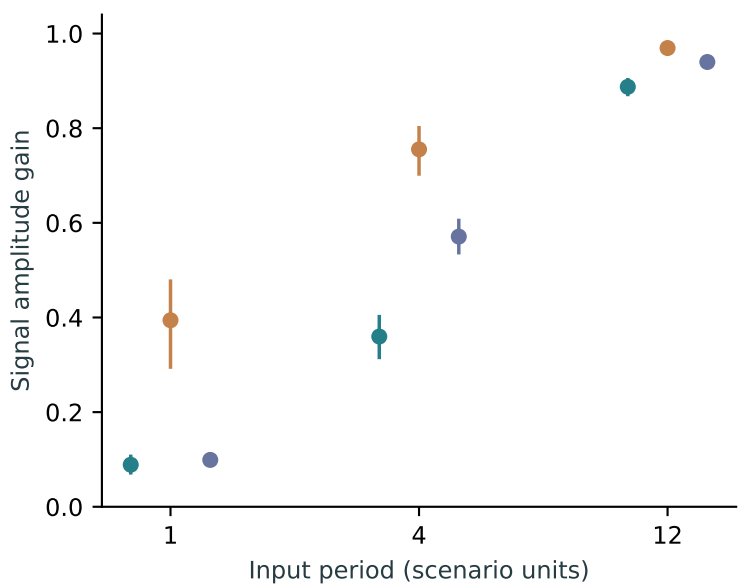
g All networks: ○ midpoint, ■ within reach



h 22 environmentally matched pairs



i Transmission depends on timescale



Distance / $\sqrt{\text{basin area}}$; time is not measured days/months. Maps retain native form with independent physical extents. Controls preserve mean delay.
g/i: HUC4 bootstrap 95%; h: paired-HUC4 95%. Five inputs/reach check spatial resolution. Conservative routing preserves load and steady concentration.